Lifting Line Group Assignment

Hide Assignment Information

Group Category

Group Assignments

Group Name

Group 8

Instructions

Do this group assignment in groups of **3**. You can make the groups yourself and/or use the Discussion board to find a group. This assignment counts for 30% of your grade.

15 h

Enroll for Group Assignments in the menu to create your group first!

Overview

In this group assignment you will create your own frozen vortex wake model of a wind turbine or a propeller, as in the case of the BEM assignment. You can find step by step instructions in the references: Katz and Plotkin [1], and the online tutorial [2].

Evaluation of a single rotor

- You will use the same geometry as for the rotor of the BEM assignment (Part 2), and analyze it at the same operating conditions as listed in the BEM assignment. Determine Cl and Cd at the 1/4 chord location, as described in [2] and in the lecture videos/powerpoint.

Wind-turbine

Baseline design of the wind turbine rotor

Rotor Specs	
Radius	50 m
# of blades	3
Blade starts at	0.2 r/R (before that circular root section without influence)
Twist	14*(1-r/R) (for r/R>0.2)
Blade pitch	-2 degrees

Chord distribution function	(3*(1-r/R)+1) m (for r/R>0.2)
Airfoil	DU 95-W-180 (polar in attachment at end of description)
Operational Specs	
U0	10 m/s
TSR (lambda)	6, 8, 10
Rotor yaw angle	0 degrees

Baseline design of the propeller rotor

Basic Rotor Specs	
Radius	0.70 m
# of blades	6
Blade Specs	
Blade starts at	0.25 r/R (before that hub)
Twist	-50*(r/R) + 35 deg (for r/R>0.25)
Collective blade pitch	46 degrees at r/R = 0.70
Chord distribution function (*c*/*R*)	0.18 – 0.06*(r/R) (for r/R>0.25)
Airfoil	ARA-D8% (polar in attachment at end of description)
Operational Specs	
U0	60 m/s
Advance ratios	1.6, 2, 2.4
ISA altitude	2000 m
Freestream incidence angle	0 degrees
Rotor yaw angle	0 degrees

The local blade pitch (defined as in Figure 1) is equal to the sum of the local blade twist angle (blade characteristic) plus the collective blade pitch at the reference location (operational characteristic). In this case, the reference location is chosen at r/R=0.70. The blade twist at this point then needs to be zero by definition.

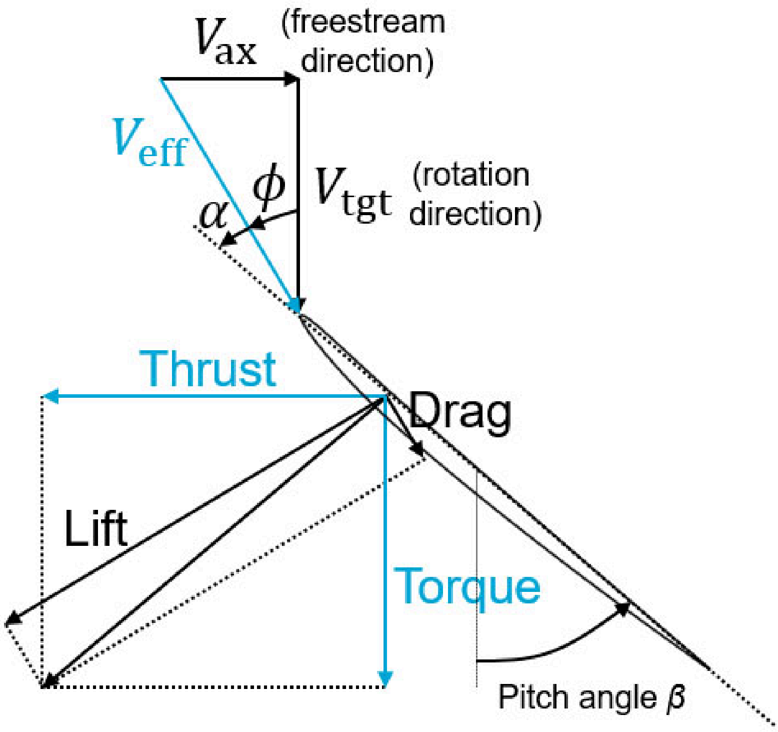

Figure 1: Definition of blade pitch for propeller case.

For this assignment you need to submit the following:

- A report, 20 pages maximum containing:
  - A SHORT introduction [5']
  - A flowchart of your code [5']
  - Main assumptions with an explanation of their impact [5']
  - Plot the non-dimensioned (explain the non-dimensioning) results of the simulation, including a discussion of the results:
    - radial distribution of the inflow angle and the angle of attack [8']
    - radial distribution of the axial and tangential/azimuthal induction [8']
    - radial distribution of the axial and tangential/azimuthal loading [8']
    - total thrust coefficient (CT) and power coefficient (CP) [8']
    - radial distribution of the circulation (discuss the generation and release of vorticity in relation to the loading and circulation over the blade) [8']
- Plots comparing the results to the results obtained in BEM [12']
- Plots with explanation of sensitivity of results to choice of:
  - assumed convection speed for frozen wake [7']
  - discretization of the blade (constant, cosine) [7']
  - azimuthal discretization (number of wake segments per rotation) [7']
  - length of the wake (number of rotations), including convergence of the solution with wake length [7']
- Include a SHORT discussion/conclusion, comparing BEM and the lifting line approach, including assumptions [5']

Note: Code in .zip or .rar file. Make sure the code is ready to run, so include all relevant files.

References

[1] Katz and Plotkin, Low Speed Aerodynamics, pp. 167-183 and 331-338

Tutorial

Use the online tutorial to support the development of your code - see link and online frame below

Online tutorial on how to program you own Lifting Line model

You can use the interactive presentation in the iframe below or click the link below to open the presentation in a new window.

Online tutorial on Lifting Line model

We blocked this for you

https://csimaoferreira.github.io/Ro...

This media is from another website. Do you want to allow it?

Allow ☐ Always trust this URL

Due on 31 May, 2026 23:59

Available on 24 April, 2026 20:34. **Access restricted before availability starts.**

Available until 08 June, 2026 23:59. **Access restricted after availability ends.**

Attachments

- polar DU95W180 (3).xlsx (11,73 KB)
- ARAD8pct_polar.txt (2,95 KB)

Download All Files

Submit Assignment

Files to submit

(0) file(s) to submit

After uploading, you must click Submit to complete the submission.

Add a File

Comments